Eco Harvesting using 5G Technology

Problem Statement

“Farmers face undetected crop diseases and rely on blanket pesticide spraying. Often pesticides are applied across entire fields even when only a few plants are infected.”

Current Challenge : Environmental impact of this problem is it wastes a lot of chemicals, increases costs, and harms soil and surrounding ecosystems through excessive chemical use

Limitations in existing solution : Traditional farming relies on **manual crop inspection**, which is slow and error-prone. Early **signs of disease can be missed**, allowing infections to spread unchecked. **This lack of real-time monitoring** leads to excessive chemical use, **reduced efficiency**, and environmental damage.

Objectives

1. Use drones and **AI image analysis (e.g., ResNet152V2) to detect plant diseases early** so farmers can intervene before damage spreads.

2. Reduce chemical overuse by applying pesticides **only to infected plants** instead of spraying entire fields.

3. Use **GPS mapping and smart-spraying systems** to target water and pesticide application precisely to affected areas.

4. Leverage **high-speed 5G for instant data processing** and real-time, actionable insights that replace manual inspection.

5. **Optimize essential resources** (water, fertilizers, pesticides) to lower costs and make farming more sustainable and eco-friendly.